Harsh Sinha

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EDUCATION

International Institute of Information Technology Bangalore **August 2023 – July 2028**
Integrated Master of Technology in Computer Science and Engineering ***[CGPA - 3.61/4.00]***

EXPERIENCE

COMET Lab **June 2026 – July 2026**
Summer Research Intern *IIIT Bangalore*

- Emulated a **9-host, 6-switch** network in **Mininet** with **6 realistic link profiles** (datacenter, backbone, home fiber, LTE, lossy Wi-Fi, satellite) by tuning bandwidth, delay, loss, and queue depth per link.
- Replaced raw UDP with **SRT** (NAK retransmissions, **3s latency buffer**) after observing severe frame corruption on lossy links, and prototyped adaptive bitrate that raised mean VMAF from **33.5 to 50.0** on the worst link.
- Instrumented per-second QoS sampling across **28 streams** using kernel **rx_bytes** counters for throughput and ping probes for latency, jitter, and loss, and merged frame-level **Netflix's VMAF** scores into a **566-row QoS-QoE dataset**.

Centre for Technology Research and Innovations for Digital Governance **May 2025 – July 2025**
Summer Intern *IIIT Bangalore*

- Extended the **Spring Boot** backend for a Citizen Complaint Portal by engineering 10+ custom **REST APIs**, with filtered queries and multi-table joins.
- Secured APIs using **Spring Security** by integrating **Google OAuth 2.0** for authentication, issuing custom JWTs post-login, and enforcing role-based access control across 4 roles via a JWT filter chain.

PROJECTS

Food-e | *Node.js (Express), MongoDB, Redis* **August 2025 – Present**

- Engineered the complete backend for a college-wide mess management platform, inaugurated by the Institute Director, serving **1,600+ registered users** with ∼**900 daily active users** and a **61% retention rate**.
- Designed role-based access across **4 roles** (Users, Admins, Committee Members, Caterer Managers), supporting features like ratings, image feedback, sick-meal requests, and PDF report generation with per-item average ratings.
- Implemented secure authentication using **JWT** (access & refresh tokens) and **Microsoft OAuth**, restricting access to verified college users, with **425+ feedback submissions** processed to date.
- Optimized performance using **Redis** caching and **Cloudinary** for file storage, with cron jobs for automated cleanup and MVC architecture for maintainability.

Inference Worker Fleet | *FastAPI, Redis, PostgreSQL* **June 2026 – Present**

- Architected a distributed inference system with a **Redis** job queue and a **multiprocessing** pool of **8 worker processes**, decoupling request handling from inference for 4 ML models with **sub-10ms** job submission latency.
- Diagnosed and fixed concurrency bottlenecks across 3 layers of the stack (CPU thread oversubscription, database connection-pool exhaustion, API thread-pool saturation), improving multi-worker throughput by **1.6x**.
- Built crash recovery via a watchdog that auto-restarts dead workers and a reaper that detects and requeues stuck jobs within **30 seconds**.

AcadOS | *C, POSIX, Linux* **May 2025 – June 2025**

- Developed a concurrent TCP client-server system in **C** using **POSIX pthreads** with a thread-per-connection model, sustaining **26,000+ RPS** under **500 concurrent clients** with sub-6ms median latency.
- Engineered file-based persistence with **POSIX semaphores** and **flock()** for atomic writes, role-based authentication with **termios** password masking, and SIGINT handling to flush sessions.
- Stress-tested across 11 concurrency levels (5–500 clients, 15,000 transactions), achieving peak **32,572 RPS** and sub-**0.34ms** p99 tail latency at low concurrency.

TECHNICAL SKILLS

Languages: C++, Java, C, Python, JavaScript
Technologies, Frameworks and Databases: Spring Boot, Node.js, Express.js, PostgreSQL, MongoDB, Redis
Relevant Coursework: Data Structures, Algorithms, Operating Systems, Database Systems, Object-Oriented Programming, Machine Learning, Programming Languages, Computer Networks

ACHIEVEMENTS

Dean's Merit List | *IIIT Bangalore* **June 2024 – Present**
Recognized for outstanding academic performance for **3 consecutive Academic Years** (2023–2026).